SRI
SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL, CHENNAI 600 063
PERIODIC TEST 1 – 2026-2027
COMPUTER SCIENCE (083)
SET 3

CLASS: XII
DATE: 10.06.2026

MAXIMUM MARKS: 20
DURATION: 40 Mins

SECTION A

20 * 1 = 20

1. statistics.mode([10,10,12,14,11,11,15,15,16,15]) will return _____
(consider module is imported)

(a) 10 (b) 15 (c) 11 (d) Error

2. A Function that does not have any return value is known as.....

(a) Library function (b) Void function (c) Fruitful function (d) None of the above

3. A function is defined with only 3 default parameters; then how many minimum number of parameters should be there in function call?

(a) one (b) two (c) three (d) zero

4. The default value for a parameter is defined in function .....

(a) Definition (b) Return statement

(c) Header (d) None of the above

5. The values being passed through a function call statements are called _____

(a) Actual parameter (b) Default parameter

(c) Formal parameter (d) None of these

6. If the return statement is not used in the function then which type of value will be returned by the function?

(a) int (b) str (c) float (d) None

7. What is the order of resolving scope in Python?

(a) BGEL (b) LEGB (c) GEBL (d) LBEG

8. What will be the output of the following code?

```
def a(b=11, c=21):
    b += 13
    c -= 13
    return b+c
print(a(25), a(35))
```

(a) 15 18 (b) 46 56 (c) 25 35 (d) 13 12

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9. Which of the following statements is not true for parameter passing to functions?

(a) You can pass positional arguments in any order.

(b) You can pass keyword arguments in any order.

(c) You can call a function with positional and keyword arguments.

(d) Positional arguments must be before keyword arguments in a function call

10. What will be the output of the Python code?

```
def testify(a,b):
    return a-b
sum=testify(22,55)
print(sum+30)
```

(a) 33 (b) -33 (c) 3 (d) -3

11. What is returned by int(math.pow(3, 2))?

(a) 6 (b) error, third argument required

(c) 9 (d) error, too many arguments

12. What will be the output of the following Python code?

```
count =1
def dothis():
    global count
    for I in (1,2,3):
        count+=1
dothis()
print (count)
```

(a) 4 (b) 1 (c) None (d) 0

13. Predict the output of the following code

```
def func1(list1):
    for x in list1:
        print(x.lower(),end="#")
func1(["New","Delhi"])
```

(a) [New, Delhi] (b) new#delhi# (c) newdelhi# (d) New#Delhi#

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14. Which of the following function header is correct?

(a) def cal_si(p=100, r, t=2)
(b) def cal_si(p, r=8, t)
(c) def cal_si(p=100, r=8, t)
(d) def cal_si(p, r=8, t=2)

15. A function code is given as follows:

```
def study (num = 5):
    print(num + 5)
```

Assertion: We can call the above function either by statement study(7) or study( ).

Reason: As the function contains default arguments, it depends on the caller that the above function can be called with or without the value.

(a) Both A and R are true and R is the correct explanation for A.

(b) Both A and R are true and R is not the correct explanation for A.

(c) A is true but R is false

(d) A is false but R is true

16. What possible output are expected to be displayed on screen at the time of execution of the program from the following code?

```
import random
Colours = ["VIOLET","INDIGO","BLUE","GREEN", "YELLOW","ORANGE","RED
End = randrange(2)+3
Begin = randrange(End)+1
for i in range(Begin,End):
    print(Colours[i],end="&")
```

(a) INDIGO&BLUE&GREEN&
(b) BLUE&GREEN&YELLOW&
(c) VIOLET&INDIGO&BLUE&
(d) GREEN&YELLOW&ORANGE&

17. Study the following program and select the possible output:

```
import random
X=random.random()
Y=random.randint(0,4)
print(int(x),":",y+int(x))
```

(a) -1:0
(b) 1:6
(c) 2:4
(d) 0:3

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18. What is the output of the program given below?

```
x = 50
def func (x) :
    x = 2
func (x)
print ('x is now', x)
```

(a) x is now 50 (b) x is now 2 (c) x is now 100 (d) Error

19. Find the output of following code:

```
x=100
def study( ):
    global x
x=50
print(x)
```

(a) 100 (b) 50 (c) Error (d) none of the above

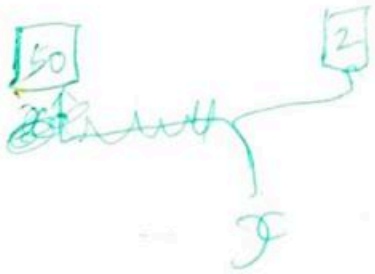
20. What will be the output of the following Python code?

```
def change(one, *two):
    print(type(two))
change(1,2,3,4)
```

(a) Integer (b) Tuple

(c) Dictionary (d) An exception is thrown

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SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL

CHENNAI - 600 063

TEST - 1 (2026 - 2027)

COMPUTER SCIENCE (083)

CLASS: XII **MAXIMUM MARKS: 35**

DATE: 20.06.2026 **DURATION: 1 Hrs. 30 mins.**

GENERAL INSTRUCTIONS:

1. Please check this question paper contains 18 questions.
2. The paper is divided into 5 Sections- A, B, C, D and E.
3. Section A, consists of 10 questions (1 to 10). Each question carries 1 Mark.
4. Section B, consists of 3 questions (11 to 13). Each question carries 2 Marks.
5. Section C, consists of 2 questions (14 & 15). Each question carries 3 Marks.
6. Section D, consists of 2 questions (16 & 17). Each question carries 4 Marks.
7. Section E, consists of 1 question (18) carries 5 Marks.
8. All programming questions are to be answered using Python Language only.

SECTION A

10 * 1 = 10

1. Evaluate the following expression:

   a) False and bool(15/5*10/2+1)

   b) float(5 + int(4.39 + 2.1) % 2)

2. Write the output for the following python snippet.

```
LST=[1,2,3,4,5,6]
for i in range(6):
    LST[i-1]=LST[i]
print(LST)
```

3. The explicit conversion of an operand to a specific type is called as _____

4. What would be the data type of variable data in the following statements?

   a) data = f.read()

   b) data = f.read(10)

   c) data = f.readline()

   d) data = f.readlines()

5. Why is it required to close a file after performing the reading and writing operations on a file?

6. Write two differences between with open() and flush() with syntax examples.

7. State True or False:

   "Tuple is datatype in Python which contain data in key-value pair."

8. What possible outputs are expected to be displayed on screen at the time of execution of the program from the following code? Also specify the maximum value that can be assigned to each of the variables L and U.

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```
import random
Arr=[10,30,40,50,70,90,100]
L=random.randrange(1,3)
U=random.randrange(3,6)
for i in range(L,U+1):
    print(Arr[i],"@",end="")
```

(i) 40 @50 @ (ii) 10 @50 @70 @90 @ (iii) 40 @50 @70 @90 @ (iv) 40 @100 @

Q9 and 10 are ASSERTION AND REASONING based questions.
Mark the correct choice as
(a) Both A and R are true and R is the correct explanation for A
(b) Both A and R are true and R is not the correct explanation for A
(c) A is True but R is False
(d) A is false but R is True

9. Assertion (A): A variable declared as global inside a function is visible with changes made to it outside the function.
   Reasoning (A): All variables declared outside are not visible inside a function till they are redeclared with global keyword.

10. Assertion (A):- If the arguments in a function call statement match the number and order of arguments as defined in the function definition, such arguments are called positional arguments.
    Reasoning (R):- During a function call, the argument list first contains default argument(s) followed by positional argument(s).

SECTION B 3 * 2 = 6

11. Write the output for the following python code:

```
def Change_text(Text):
    T=""
    for K in range(len(Text)):
        if Text[K].isupper():
            T=T+Text[K].lower()
        elif K%2==0:
            T=T+Text[K].upper()
        else:
            T=T+T[K-1]
    print(T)
Text= "LeaARN CODIng"
Change_text(Text)
```

12. Define tell() and seek() and Illustrate it with example.

13.(a) Given is a Python string declaration: str="CBSE Examination@2026"
Write the output of: print(str[-1:-15:-2])

(b) Write the output of the code given below:

```
d = {"name": "Akash", "age": 16}
d['age'] = 27
d['city'] = "New Delhi"
print(d.items())
```


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SECTION C

2 * 3 = 6

14. Write a method/function COUNTWORDS_S() in Python to read the content from a text file DATA.TXT. Display those sentences which start with the letter 'S', and also contain the word: "the". Along with each sentence, display the total number of words it contains as 'Word Count'.

For Example: If DATA.TXT consists of:

"Always chase your dreams and never look back. Success comes to those who work hard. Stars shine brightest in the darkest night. All the efforts you put in today will reward you tomorrow. Smile through the pain."

The Output will be:

Stars shine brightest in the darkest night. Word Count: 7

Smile through the pain. Word Count: 4

(Or)

Write a function/method named search() in Python to read data from a text file named 'Persons.txt' and display the name and address of every individual whose age is strictly greater than 35.

Let the file "Persons.txt" include the following sample text:

Samyukta, Mumbai, 35

Aniket, Hyderabad, 42

Sarth, Bangalore, 31

The Output will be:

Aniket, Hyderabad

15. a) Write a function lenFOURword(L), where L is the list of elements (list of words) passed as argument to the function. The function returns another list named 'indexList' that stores the indices of all four lettered word of L.

For example:

If L contains ["DINESH", "RAMESH", "AMAN", "SURESH", "KARN"]

The indexList will have [2, 4] **(2)**

b) Identify the valid and invalid function calls for the following function and state the reason why it is invalid? **(1)**

```
def Student(rollno, name, section='A'):
    print(rollno, name, section)
```

(i) Student(101,"Arun")

(ii) Student(name="Riya",102)

(iii) Student(103,name="Deepa",section='C')

(iv) Student(104,"Kavi","B")

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SECTION D

2 * 4 = 8

16. Fill in the blanks for the code snippet below and display the final output.

INFO.TXT
Small steps build long-term success.
Difficult challenges reveal hidden strengths.
Believe in your unique potential.
Daily habits create real progress.
Let curiosity guide your journey

```
f = open("INFO.TXT", "r")
pos = _____                  #1: Get current byte position
content = _____              #2: Read all lines into a Python list
print(_____)                 #3: Displays the third line of the file
_____                        #4: Move the file pointer back to the start
print(_____)                 #5: Display total number of lines
_____                        #6: Free up system resources by disconnecting the file
```

17. Complete the following program by filling in the blanks to create a function MODIFY(LST) that accepts a numeric list as an argument. The function should calculate the sum of all original elements in a global variable, double each element inside the list itself, and return the modified list. Display the final output

```
TOTAL = _____
def MODIFY(LST):
    global _____
    for I in range(_____):
        TOTAL = TOTAL + LST[I]
        LST[I] = LST[I] * _____
    return _____
NUM = [10, 20, 30, 40]
RESULT = _____(NUM)
print("output =", _____)
print("Sum =", _____)
```

SECTION E

1 * 5 = 5

18) a) Write a user defined function called WORDLEN() which accepts a list of words as parameter. The function should create a dictionary where the key is the length of the word and the value is a list of words having the same length. Print the final dictionary.

Example:

Input: ["Pen", "Book", "Cat", "Apple"]

Output: {3:['Pen', 'Cat'], 4:['Book'], 5:['Apple']} (3)

b) Explain types of arguments with examples. (2)

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Roll No.: | | | | | | |

SRI
SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL
CHENNAI 600 063
TEST 2 – (2026 - 2027)
COMPUTER SCIENCE

CLASS: XII
DATE: 22.07.2026

MAXIMUM MARKS: 35
DURATION: 1 Hrs. 30 mins.

GENERAL INSTRUCTIONS:

1. Please check this question paper contains 18 questions.
2. The paper is divided into 5 Sections- A, B, C, D and E.
3. Section A, consists of 10 questions (1 to 10). Each question carries 1 Mark.
4. Section B, consists of 3 questions (11 to 13). Each question carries 2 Marks.
5. Section C, consists of 2 questions (14 & 15). Each question carries 3 Marks.
6. Section D, consists of 2 questions (16 & 17). Each question carries 4 Marks.
7. Section E, consists of 1 question (18) carries 5 Marks.
8. All programming questions are to be answered using Python Language only.

SECTION A

10 * 1 = 10

1. Consider the following operation performed on a stack of size 5.
   Push(1);Pop();Push(2);Push(3);Pop();Push(4);Pop();Pop();Push(5);
   After the completion of all operation, the number of elements present in stack are
   a) 1 b) 2 c) 3 d) 4
2. Consider the statements given below and predict the ouput:

```
def fun2():
    global x
    print(x, end=":")
    x=500
    print(x, end=":")
    x=x+1
x=5
print(x, end=":")
fun2()
print(x,end=":")
```

3. State True or False: Processing of text files is faster than binary files.
4. Write the statement that will move file pointer 10 bytes backward from current position.
5. Name any 2 applications of stack.
6. Which of the following function definition header is wrong?
   a) def sum(n1,n2,n=3): b) def sum(p1,p2=4,p3=5):
   c) def div(p1=4,p2,p3): d) def mul(p1,n1,m1):
7. Write the output of the following:

```
>>>f=open('test.txt', 'w')
>>>f.write("File\n#Handling")
```

   a)17 b)16 c)14 d)15
8. In a stack, if a user tries to remove an element from empty stack it is called_____

Q9 and 10 are ASSERTION AND REASONING based questions.
Mark the correct choice as
(a) Both A and R are true and R is the correct explanation for A
(b) Both A and R are true and R is not the correct explanation for A
(c) A is true but R is false.
(d) A is false but R is true.

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9. Assertion (A): No matter what exception occurs, you can always make sure that some common action takes place for all types of exceptions.
Reason (R): The finally block contains the code that must execute.

10. Assertion (A): A function may or may not return a value when called. The return statement returns the values from the function.
Reason (R): The return statement does the following:
- returns the control to the calling function.
- return value(s) or None.

3 * 2 = 6

SECTION B

11. Predict the output of the Python code Given below: (2)

```
def fun_para(x=5,y=10,z=1005):
    z=x/2
    res=y//x+z
    return res
a,b,c=20,10,1509
print(fun_para(),fun_para(b),sep='#')
res=fun_para(10,20,6015)
print(res, "@")
print(fun_para(z=999,y=b,x=5))
print(fun_para(60))
```

12.a) Predict the output (1)

```
def myfunc(x = None) :
    result = ""
    if x is None:
        result = "No argument given"
    elif x == 0:
        result = "Zero"
    elif 0 < x <= 3:
        result = "x is between 0 and 3"
    else:
        result = "x is more than 3"
    return result
print(myfunc(3.5))
print(myfunc())
print(myfunc(0))
print(myfunc(1))
```

b) List the situation(s) in which the following errors occur: Illustrate with example code
(i) ValueError (ii) TypeError (1)

13. Explain the scope of variables with examples. (2)

SECTION C

2 * 3 = 6

14. a) Write a function, WordCount() in Python to read a text file, "story.txt", and display those words in it that begin with consonants and end with vowels. Also, the function will count those words and return how many such numbers are in the same text file. (2)

b) A text file song.txt contains the following contents in it : (1)

Life goes on as it never ends

What will be the output of the following code snippet?

```
f1=open("song.txt","r")
s1=f1.read(5)
s2=f1.readline(4)
s3=f1.read(3)
print(s1,s3,sep="#")
```


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15.a) Two lists Lname and Lage contain name of person and age of person respectively. A list named Lnameage is empty. Write functions as per details given below:
Push_na(): It will push the tuple containing pair of name and age from Lname and Lage whose age is above 50.
Pop_na(): It will remove the last pair of name and age and also print name and age of the removed person. It should also print "underflow" if there is nothing to remove. (2)

For example, the two lists have the following data:
Lname=['Narender', 'Jaya', 'Raju', 'Ramesh', 'Amit', 'Piyush']
Lage= [45, 23, 59, 34, 51, 43]
After Push_na() the Lnameage Stack contains:
[('Raju', 59), ('Amit', 51)]
The output of first execution of Pop_na() is:
The name removed is Amit
The age of person is 51

b) What is Data Structure? List out the operations can be performed in it. (1)

SECTION D

2 * 4 = 8

16. Mr. Snehant is a software engineer working at TCS. He has been assigned to develop code for stock management; he has to create a CSV file named stock.csv to store the stock details of different products. The structure of stock.csv is : [stockno, sname, price, qty], where stockno is the stock serial number (int), sname is the stock name (string), price is stock price (float) and qty is quantity of stock(int).
Mr. Snehant wants to maintain the stock data properly, for which he wants to write the following user-defined functions:
AcceptStock() – to accept a record from the user and append it to the file stock.csv. The column headings should also be added on top of the csv file. The number of records to be entered until the user chooses 'Y' / 'Yes'.
StockReport() – to read and display the stockno, stock name, price, qty and value of each stock as **price*qty**. As a Python expert, help him complete the task. (4)

17.a) Write a function seq_to_dict(ls, tp) in Python that takes one list and one tuple as two arguments. The function will return a dictionary with an update of key-value pairs with keys from a list and values from a tuple.For example, Consider the following:
List ls=[1,2,3,4] Tuple tp=("Pranay", "Shuvam", "Snehant","Swasat")) (2)
The returned dictionary will be: {1: 'Pranay', 2: 'Shuvam', 3: 'Snehant', 4: 'Swasat'}
b) Write a code snippet that will create an object called fileout for writing; associate it with the filename 'STRS'. The code should keep on writing strings to it as long as the user wants. (2)

SECTION E

1 * 5 = 5

18. (i) What are the similarities and difference between a+ and w file modes in Python. (1)

(ii) A binary file "product.dat" has structure [PNO, PNAME, BRAND_NAME, PRICE] to maintain and manipulate Product details. (4)

- Write a user defined function CreateFile() to input data for a record and add to "product.dat" file. The number of records to be entered until the user chooses 'Y'
- Write a function CountData(BRAND_NAME) in Python which accepts the brand name as parameter and count and return number of products by the given brand are stored in the binary file "product.dat".

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SRI

SRIMATHI SUNDARAVALLI MEMORIAL SCHOOL

CHENNAI 600 063

I TERM EXAMINATION - (2026 - 2027)

COMPUTER SCIENCE

CLASS: XII **MAX. MARKS: 70**

DATE: 15.09.2026 **DURATION: 3 Hrs.**

GENERAL INSTRUCTIONS:

1. Please check this question paper contains 37 questions.
2. The paper is divided into 5 Sections- A, B, C, D and E.
3. Section A, consists of 21 questions (1 to 21). Each question carries 1 Mark.
4. Section B, consists of 7 questions (22 to 28). Each question carries 2 Marks.
5. Section C, consists of 3 questions (29 to 31). Each question carries 3 Marks.
6. Section D, consists of 4 questions (32 to 35). Each question carries 4 Marks.
7. Section E, consists of 2 question (36 & 37) carries 5 Marks.
8. All programming questions are to be answered using Python Language only.

SECTION A **21 * 1 = 21**

1. What is the value of the expression:

   a) 16 // 3 + 4 ** 2 b) float(5 + int(4.39 + 2.1) % 2)

2. Predict the output:

```
S = "Welcome 2026"
print(S[-4:].isdigit())
```

3. Predict the output:

```
stack = {"x": 10, "y": 20, "z": 30}
val_1 = stack.popitem()
val_2 = stack.pop("x")
val_3 = stack.popitem()
print(val_1, val_2, val_3)
print(stack)
```

4. Predict the output:

```
def check(x, y=2):
    return x ** y
print(check(3) + check(2, 3))
```

5. Predict the output:

```
data = {"apple": 5, "banana": 3}
result_1 = data.get("cherry")
result_2 = data.get("orange", 0)
print(result_1, result_2)
```

6. a) What is the minimum possible value generated by random.randint(3, 7)?

   b) Predict the output:

```
import math
print(math.floor(-3.4))
```


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7. Identify the type of error in the following:

```
import random
val = random.rand_num(1, 10)
```

8. Which block in Python exception handling executes no matter whether an exception occurs or not?

(a) except (b) else (c) finally (d) try

9. What will be the output of the following program?

```
a = {'A':10,'B':20} ;b = {'B':20, 'A':10} ;print(a==b and a is b)
```

10. What will be the output of the following python statement?

```
s = "HOME ALONE"
p = s.split("O",1)
print(p)
print(p[1][:2]+p[-1])
```

11. What will be the output of the following string operation?

```
"MALAYALAM".partition("MA")
```

12. Which of the following statements will generate an error?why?

```
st = "PYTHON"
t = st*5          Statement(1)
u = st[0] + "M"   Statement(2)
st[0] = "K"       Statement(3)
st = st + st      Statement(4)
```

13. What will be the output of the following python statement?

```
s = "MONGO"
print(sorted(s))
```

14. What will be the value of abs(3 – abs(-10))?

(a) 13 (b) -13 (c) –7 (d) 7

15. Sushma gets the current date and time as a string x. Its value is "2021-10-30 12:49:44.216062". Which of these will contain the date in yyyy-mm-dd format?

(a) x[0:9] (b) x[0:10] (c) x[1:10] (d) x[1:11]

16. What will be the output of the following Python code?

```
int('65.43')
```

17. Which of the following is NOT a standard built-in exception in Python?

(a) ValueError (b) IndexError (c) NullPointerException (d) KeyError

18. _____ Exception is raised when Esc, Del or Ctrl + C is pressed during program execution and normal program flow gets disturbed.

19. State True or False: "In Python, can a function return multiple values simultaneously. 25

Q20 and 21 are ASSERTION AND REASONING based questions.

Based on the statements given below, select one of the following option.

(a) Both A and R are true, and Reason is the correct explanation for Assertion

(b) Both A and R are true, and Reason is not the correct explanation for Assertion

(c) Assertion is True but Reason is False

(d) Assertion is false but Reason is True

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20. Assertion : When passing a mutable sequence as an argument, function modifies the original copy of the sequence.
Reason : Function can alter mutable sequences passes to it.

21 Assertion: Python is said to have broadly two types of files – binary and text files, even when there are CSV and TSV files also.
Reason: The CSV and TSV are types of delimited text only where the delimiters are common and tab respectively.

SECTION B

7 * 2 = 14

22. Rewrite the following Python code after removing all syntax error(s). Underline each correction made.

```
Define Calc_Interest( Principal, Rate = 5.5 )
   Amount = (Principal * Rate * Time) / 100
   return Amount
P = 10000
Final = calc_interest(P)
print("Interest is ", Final)
```

23. Suggest the maximum and minimum possible values that can be assigned to the variable Go in the code below. Also, choose the correct output(s) from the given options (i) to (iv).

```
import random
POINTS = [10, 20, 30, 40, 50, 60]
Low = random.randint(1, 3)
High = random.randint(2, 4)
for Go in range(Low, High + 1):
   print(POINTS[Go], end="#")
```

(i) 10#20#30# (ii) 20#30#40#50# (iii) 30#40#50#60# (iv) 20#30#40#

24. Find and write the output of the following Python code segment:
Assume the text file WISH.TXT contains the string: WELCOME TO PYTHON

```
file = open("WISH.TXT", "r")
data1 = file.read(7)
print(data1)
print(file.tell())
file.seek(11)
data2 = file.read(6)
print(data2)
file.close()
```

5. Write a Python statement to perform the following tasks:
a) Add a new key Grade with the value "A" to the dictionary student
b) Find the number of key-value pairs present in the dictionary student
c) Remove the key "Salary" from the dictionary employee
d) Check whether the string password contains only digits

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26. What is the output of following code fragments.

```
def increment(n):
    n.append([49])
    return n[0], n[1], n[2], n[3]
L = [23, 35, 47]
m1, m2, m3, m4 = increment(L)
print(L)
print(m1, m2, m3, m4)
print(L[3] == m4)
```

27. Find and write the output of the following Python code:

```
Num = 10
def Process(Sep):
    global Num
    Num += 5
    print(Num, end = Sep)
    Num = 40
    print(Num, end = Sep)
print(Num, end = "#")
Process("@")
print(Num)
```

28. Write a function in Shift(Lst), Which accept a List 'Lst' as argument and swaps the elements of every even location with its odd location and store in different list
For Example: if the array initially contains [2, 4, 1, 6, 5, 7, 9, 2, 3, 10]
then it should contain [4, 2, 6, 1, 7, 5, 2, 9, 10, 3]

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SECTION C

3 * 3 = 9

29. A gaming application tracks a scoreboard stack named PlayerStack where each element is a Dictionary containing player metrics: Key as Player_ID and values as [Game_Name,Score]. Write a Python program containing:
    1. push_player(player_id, game, score): To collect parameters, bundle them into a player dictionary, and push it onto PlayerStack.
    2. pop_player(): To pop the top player record from the stack. If the player's Score key is greater than 500, print "High Scorer Popped!" along with their ID; otherwise, just print their details. Handle underflow safely.
    3. peek_player(): To print the Player_ID of the person currently occupying the top position of the stack.

30. Write a function filter_copy() in Python to read from a text file DIARY.TXT and copy al the words that have a length of less than 4 characters into another text file name SHORT.TXT

31. Write a function alter_string_list(STR_LIST) in Python that accepts a list of strings. The function should modify each string inside the list: if the string length is odd, convert it to uppercase; if the string length is even, reverse the string.
Example: If the list is ["Python", "css"], it should modify the list to ["nohtyP", "CSS"].

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SECTION D

4 * 4 = 16

food delivery application maintains its active menu in a CSV file named menu.csv th the following fields:[Item_Code, Item_Name, Category, Availability]
rite a Python program using the following functions:

. mark_out_of_stock(code) – To search for a dish using its Item_Code and change its Availability status to 'N'.

2. count_available() – To count and display the total number of dishes whose Availability status is 'Y'.

A warehouse management system maintains a stack of shipping parcels using a List structure named ParcelStack. Each element in the stack is a single tracking string code (e.g., "PKG-101", "URG-202"). Write a Python program with the following user-defined functions:

1. push_parcel(parcel_code): To push a parcel onto the stack. However, the parcel should only be pushed if the parcel_code starts with the letters "URG" (Urgent).
2. pop_parcel(): To delete and return the top-most parcel code from ParcelStack. If the stack is empty, display an "Underflow" message.
3. peek_parcel(): To view and print the top parcel code without removing it. Display "Stack Empty" if applicable.
4. display_parcels(): To display all parcel codes present in the stack from top to bottom

34. (a) Differentiate between DDL and DML statements. Provide two command examples for each category. (2)

(b) A database administrator wants to change the data type of an existing column Phone_No from INT to VARCHAR(15) and also remove an unwanted record where Status = 'Inactive'. Name the exact SQL keywords/commands required to execute both operations. (1)

(c) Difference between candidate key and primary key. (1)

35. Write a Python program to store the following dictionary,
D = {101: "Arun",102: "Bala",103: "Chitra", 104: "Divya"} in a binary file named student.dat. The program should perform the following two operations:
Read_rec() – Read and display all the records from the binary file.
Update_rec() – Accept a roll number from the user and update the corresponding student's name in the dictionary. Store the updated dictionary back into the binary file and also display if the record is not found.

SECTION E

2 * 5 = 5

36. A school tracks its sports club enrollments in a binary file named sports.dat. Each student record is saved as a List in the format [Admission_No, Student_Name, Game, Age]. Write a program that implements these 5 functions: (5)

1. create_club(): Initializes sports.dat with an initial set of 3 student list records.
2. append_student(): Prompts the user for data, and add to the file.
3. search_game(game_name): Searches the file and displays the details of all student lists matching the Game as 'Cricket'.
4. update_age(adm_no, new_age): Get Admission_No and updates its Age value with new_age.
5. remove_student(adm_no): Permanently deletes the student records matching Admission_No.

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37. a) A school library has a rule that a book can only be issued to a registered member. Each book has a unique Accession_No, and each student has a unique Admission_No. The library assistant maintains the following temporary log sheet: (3+2)

Table: Book_Issue

Accession_No	Book_Title	Admission_No	Student_Name	Return_Status
B101	Core Python	4412	Aarav Sharma	Pending
B102	Learning SQL	NULL	Unknown	Available
B103	Data Structures	4501	Meera Nair	Pending
B103	Computer Networks	4982	John Doe	Pending
B104	Wings of Fire	4412	Aarav Sharma	Pending

Answer the following based on the database rules:

1. Book B102 is currently on the shelf and has not been issued to anyone. Is it structurally correct to keep this row in the Book_Issue table with Admission_No as NULL? Explain with respect to the table's purpose.
2. The log shows Accession_No B103 assigned to two different students at the same time. Which property of a relational database is violated here? Which key constraint can prevent this?
3. Student Aarav Sharma (Admission_No 4412) has borrowed two different books (B101 and B104). Does this violate any primary key rules if Accession_No is chosen as the Primary Key?

b) Define the following database terms :

1.Tuple
2.Attribute
3.Relation
4.Domain

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